BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Hardware

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is the role of a server in a client-server network?

2. When does iterative testing happen?

A) Only after delivery
B) Only before any coding
C) During development
D) Only during installation

3. For valid integer ages 13-18, give an invalid input of the correct type.

4. A public booking form takes text and numbers. "Defensive design assumes every input is valid and safe." Correct this. What could happen?

Today's objectives

Describe the roles of a NIC, switch, router and WAP. Distinguish local MAC-based forwarding from routing between networks.

Key vocabulary - Read, cover, recall

NIC
Network Interface Card; connects a device to a network.

Switch
Connects LAN devices and forwards traffic using MAC addresses.

Router
Forwards data between networks using IP addresses.

WAP
Wireless Access Point; provides wireless connection to a network.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

NIC	A Network Interface Card (NIC) lets a computer connect to a network and transmit/receive data. Interfaces have MAC addresses.
Switch	A switch connects devices in one LAN and forwards local data using MAC addresses.
Router	A router moves data between different networks, using destination IP addresses.
Wireless access point	A WAP connects wireless devices to a network, normally through the existing wired LAN.

I Do - One worked example

A student sends coursework to a printer on the same school LAN. Explain what the NIC and switch do. [4 marks]

1. The computer and printer are in the same LAN; the data must leave the computer.

2. A NIC provides network connectivity [1] and sends the data [1].

3. The switch connects LAN devices [1] and forwards the traffic using MAC addresses [1].

4. Two distinct devices with two accurate marking points each.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A pupil visits a website outside the school network. Explain the roles of the NIC and router. [4 marks]

2. Tablets must access a school's existing wired network. Explain why a WAP and switch are needed. [4 marks]

3. Explain two differences between a router and a switch. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: What prints when local is false?

```
local = false
if local then
print("Stay on LAN")
else
print("Use router")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State what a NIC allows a computer to do. [1 mark]

Q3. A supermarket introduces wireless barcode scanners that must access its existing wired LAN. Explain the role of a WAP and a switch. [4 marks]

Q4. A museum information kiosk requests a catalogue page hosted on another network. Explain the roles of its NIC, the local switch and the router. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A router forwards between networks using IP addresses. ☐ T ☐ F
2. A Network Interface Card (NIC) lets a computer connect to a network and transmit/receive data ☐ T ☐ F
3. A switch connects devices in one LAN and forwards local data using MAC addresses. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It provides centrally managed services or resources requested by clients. ☐

Q2 C - During development ☐

Q3 12 is an integer but outside the valid range, so it must be rejected. ☐

Q4 Plan for invalid, unexpected or harmful input and handle it safely. A known user can still enter unsuitable data. ☐

Code First - Model

```
local = false
if local then
print("Stay on LAN")
else
print("Use router")
endif
```

Output: Use router

We Do - Check the model

1. The NIC connects the pupil's computer to the LAN [1] and sends/receives traffic [1]. The router forwards data between the LAN and external networks [1] using IP addresses [1]. ☐

2. WAP lets tablets join wirelessly [1] by radio [1]. Switch links the WAP to the LAN [1] and forwards data to intended LAN devices [1]. ☐

3. A switch handles traffic within a LAN [1] using MAC addresses [1]. A router forwards between networks [1] using IP addresses [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Connect to a network to send/receive data [1]. ☐

Q3 WAP connects wireless devices [1] using radio [1]. Switch links wired devices/WAP in a LAN [1] and forwards using MAC addresses [1]. ☐

Q4 NIC connects device [1] and transmits data [1]; switch connects LAN devices [1] using MAC forwarding [1]; router connects networks [1] using IP addressing [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: